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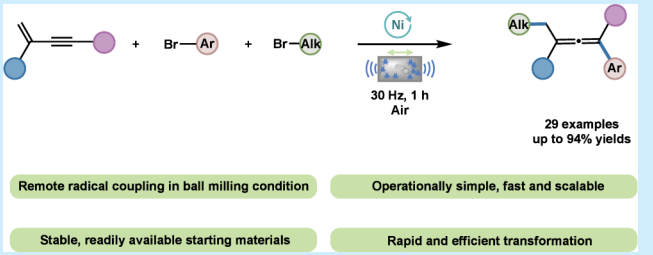
Ball-Milling-Enabled Nickel-Catalyzed Reductive 1,4-Alkylarylation of 1,3-Enynes under an Air Atmosphere

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Cite This: *Org. Lett.* 2024, 26, 7688−7694 Read Online

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ABSTRACT: A ball-mill-enabled nickel-catalyzed 1,4-alkylarylation of 1,3-enynes with organic bromides has been developed, offering a versatile method for assembling tetrasubstituted allenes. This approach, the first of ball-milling-based remote radical coupling, overcomes the limitations of traditional solution-phase methods, such as the need for air- and moisture-sensitive reagents, the use of bulk solvents, and prolonged reaction times. Given the outstanding performance of ball-milling-based radical reduction coupling reactions, we anticipate further advancements in sustainable and efficient synthetic methodologies.



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Mechanochemistry, which involves the direct absorption and utilization of mechanical energy to drive chemical reactions, has emerged as a powerful strategy in material science,[1] polymer chemistry,[2] and both inorganic and organic synthesis.[3] Its primary advantages include simplicity of operation, significant reduction in reaction times, elimination of the extensive use of harmful organic solvents, and precise energy control. This method is particularly suitable for substrates with poor solubility and allows for air- and moisture-sensitive reactions to be conducted under ambient conditions. Ball-milling conditions can be applied to a wide range of applications, including Buchwald−Hartwig,[3e,i,4] Negishi coupling,[5] Friedel−Crafts,[6] olefin metathesis,[7] Mizoroki−Heck,[8] Sonogashira coupling,[9] Suzuki−Miyaura,[10] C−H activation,[11] and radical approaches.[3b,12] These methods often achieve better reaction efficiency compared to traditional solution-phase reactions, presenting exciting possibilities for exploring previously uncharted areas of chemical space.

Allenes, which are cumulative dienes, have garnered notable attention in pharmaceutical science and synthetic chemistry due to their distinct biological activity and chemical reactivity.[13] However, classic methods for allene synthesis typically suffer from several limitations: they involve labor-intensive steps for preparing activated substrates and require stringent reaction conditions, especially for synthesizing multisubstituted allenes.[14] In this context, the 1,4-functionalization of readily accessible 1,3-enynes offers an efficient and practical synthetic approach for producing multisubstituted allenes. Although the dicarbofunctionalization of 1,3-enynes can directly form two C−C bonds to obtain multisubstituted allenes, the polar approach via allenyl anion species usually requires organolithium or organozinc reagents, which are constrained by functional group compatibility and necessitate cryogenic cooling (Figure 1A).[15] In contrast, the radical

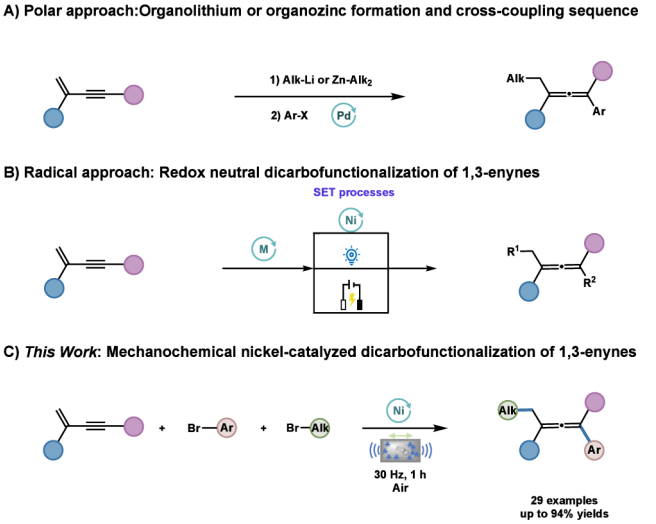


Figure 1. Overview of catalytic difunctionalization of 1,3-enynes and our approach.

approach has made significant progress in the 1,4-dicarbofunctionalization of 1,3-enynes by combining single-electron transfer processes with transition-metal catalysis.[16] Recent advancements in the dicarbofunctionalization of 1,3-enynes have employed not only nickel catalysis but also photo-

Received: July 30, 2024
Revised: August 26, 2024
Accepted: August 28, 2024
Published: August 29, 2024



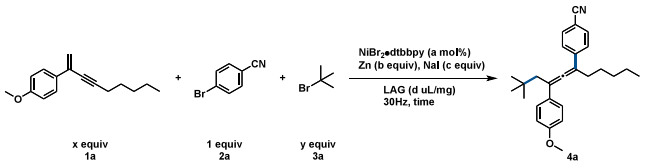
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https://doi.org/10.1021/acs.orglett.4c02729
Org. Lett. 2024, 26, 7688−7694

chemistry and electrochemistry to achieve single-electron transfer (SET) processes (Figure 1B).[17] Despite these advancements, traditional solution-phase reactions still face challenges such as sensitivity to air and moisture, long reaction durations, the need for large amounts of solvents, and a limited range of coupling partners (typically relying on alkyl iodides as radical donors). With the growing importance of sustainability in both societal and industrial processes, there is an increasing need to reduce the introduction of excessive harmful organic waste, such as solvents and chemical stoichiometric waste, while achieving efficient, widely applicable, and stable transformations. In this context, our previous work has demonstrated that nickel-catalyzed reductive cross-coupling reactions can be successfully performed using mechanochemistry.[18] Herein, we present the first example of 1,4-alkylarylation of 1,3-enynes by combining nickel catalysis with mechanical synthesis, which is also the first case of ball-milling-based remote radical coupling (Figure 1C). The high vibration frequency (30 Hz, MM500) is crucial for achieving efficient transformation, emphasizing the importance of mechanical grinding in effectively dispersing substrates, reagents, and catalysts. Notably, this process does not require inert gas or dry organic solvents, and the entire procedure can be conducted in air without special precautions or techniques.[19] This discovery broadens the substrate scope to include commercially available alkyl and aryl bromides, yielding 1,4-arylalkylated products with good functional group compatibility.

All mechanochemical reactions were conducted in a Retsch MM500 mill (stainless-steel milling jar; 30 Hz; stainless-steel balls). As a proof of concept, we initiated the exploration of the reaction using 1,3-enyne (**1a**) and selected commercially available 4-bromobenzonitrile (**2a**) and *tert*-butyl bromide (**3a**) as model substrates. These were added, along with other additives, to a 1.5 mL stainless-steel milling jar containing one grinding ball (stainless, diameter: 5.0 mm) inside a glovebox. Following simple filtration and NMR analysis, the reaction conditions yielded tetrasubstituted allene **4a** in 70% ^{1}H NMR yield (Table 1, entry 1). Increasing the quantity of 1,3-enyne (**1a**) from 1 equivalent to 1.5 equivalents improved the yield of the desired allene **4a** to 99% ^{1}H NMR yield and 94% isolated yield (entry 2). Reducing the amount of various components, such as liquid-assisted grinding agent (LAG) (entries 3, 4), *tert*-butyl bromide (**3a**, entries 5, 6), nickel salt $NiBr_2$•dtbbpy (entry 7), reductant Zn (entry 8), and additive NaI (entries 9, 10), led to decreased reaction yields. Notably, the absence of tetrahydrofuran (THF) as LAG resulted in no detectable allene **4a**, underscoring the importance of LAG in promoting uniform grinding (entry 4). Replacing THF with CH_3CN as LAG also resulted in a reduced yield (entry 11). When the reaction was conducted in air, the isolated yield of allene **4a** decreased, and TLC indicated that a portion of 4-bromobenzonitrile (**2a**) remained unreacted (entry 12). Neither increasing the amount of LAG nor extending the reaction time significantly enhanced the isolated yield of allene **4a** (entries 13, 14), but adjusting both conditions simultaneously resulted in a yield comparable to that of entry 3 (entry 15). In contrast, solution-based reactions (entries 16, 17) did not produce the desired products. Increasing the temperature from room temperature to 50 °C under the solution conditions of entry 17 only resulted in a 33% yield (entry 18). This further highlights the superiority of our mechanochemical approach for achieving efficient and practical conversion.

Table 1. Optimization of Reaction Conditions



Entry[a]	x	y	$NiBr_2$•dtbbpy (a)	b	c	LAG (d)	Time (h)	Yield (%)[b]
1	1	2.5	10	2	3	THF (0.2)	0.5	70
2	1.5	2.5	10	2	3	THF (0.2)	0.5	99(94)
3	1.5	2.5	10	2	3	THF (0.1)	0.5	69
4	1.5	2.5	10	2	3	No	0.5	0
5	1.5	2	10	2	3	THF (0.2)	0.5	58
6	1.5	1.5	10	2	3	THF (0.2)	0.5	48
7	1.5	2.5	5	2	3	THF (0.2)	0.5	82
8	1.5	2.5	10	1	3	THF (0.2)	0.5	78
9	1.5	2.5	10	2	2	THF (0.2)	0.5	86
10	1.5	2.5	10	2	1	THF (0.2)	0.5	76
11	1.5	2.5	10	2	3	CH_3CN (0.2)	0.5	80
12[c]	1.5	2.5	10	2	3	THF (0.2)	0.5	(84)
13[c]	1.5	2.5	10	2	3	THF (0.4)	0.5	(87)
14[c]	1.5	2.5	10	2	3	THF (0.2)	1	(88)
15[c]	1.5	2.5	10	2	3	THF (0.4)	1	99(93)
16[c,d]	1.5	2.5	10	2	3	THF (0.4)	12	0
17[c,e]	1.5	2.5	10	2	3	THF	12	0
18[c,f]	1.5	2.5	10	2	3	THF	12	33

[a]The reactions were carried out with 0.2 mmol of 4-bromobenzonitrile in a stainless-steel ball-milling jar (1.5 mL) with a stainless-steel ball (5 mm). [b]Yields determined by ^{1}H NMR analysis using mesitylene as an internal standard; the isolated yield is shown in parentheses. [c]Air atmosphere. [d]The reaction were carried out in a 2 mL glass bottle at room temperature for 12 h. [e]The reactions were carried out with 1 mL of THF in a 2 mL glass bottle at room temperature for 12 h. [f]The reactions were carried out with 1 mL of THF in a 2 mL glass bottle at 50 °C in an oil bath for 12 h.

With the optimized reaction conditions established, we initially explored the scopes for the 1,4-alkylarylation reaction (Figure 2). For the alkene C2-aryl substituent, aryls with electron-donating, electron-neutral, and electron-withdrawing groups exhibited excellent tolerance to the reductive 1,4-alkylarylation reaction, yielding the corresponding tetrasubstituted allene (**4a**–**4j**). The yields ranged from good to excellent (64–93%), regardless of the electronic nature of the aryl substituents of the substrate **1**. As described in the literature,[17a] this result indicates that it involves a radical intermediate rather than anionic or cationic ones. This finding, which indicates that the reaction is relatively insensitive to substituent electronic effects, provides important insights for the reaction mechanism and brings greater flexibility and broader applicability in different conditions. Remarkably, polyaromatic substrates, often deemed challenging substrates in solution-based conditions due to their limited solubility, efficiently formed the desired products (**4h** and **4i**) in 87% and 81% yield, respectively. This provides further confirmation of the advantages that the ball-milling reaction holds over the traditional solution reaction. When the linear alkane on R^2 of 1,3-enyne (**4a**, **4l**) was replaced with sterically hindered groups (**4k**, **4m**, and **4n**), the resulting allene products were obtained with relatively lower conversion rates, presumably due to steric hindrance. Next, we evaluated the scope of aryl bromides in

https://doi.org/10.1021/acs.orglett.4c02729
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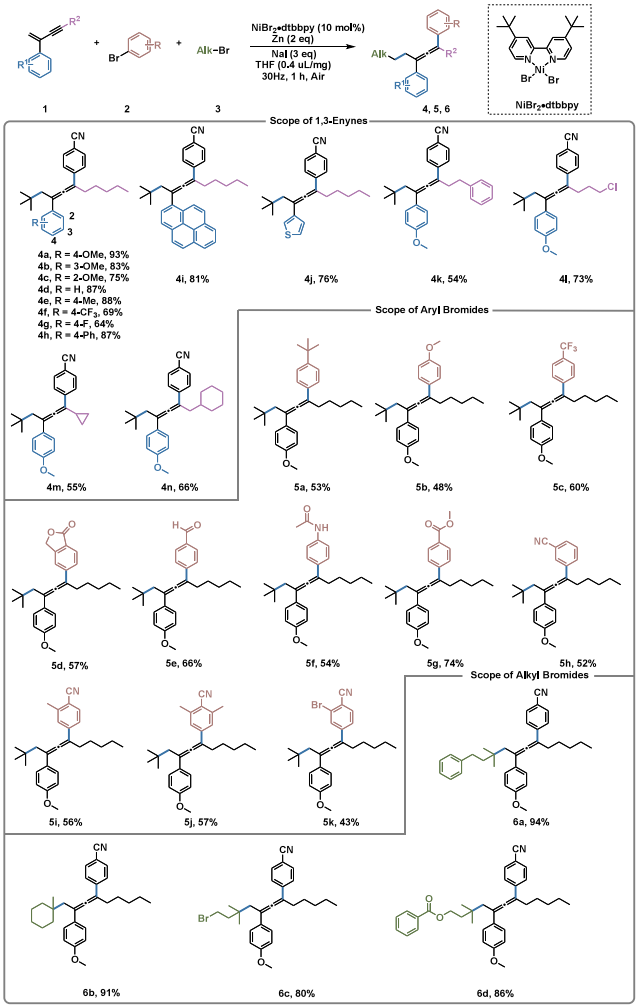


Figure 2. Substrate scope of 1,3-enynes, aryl bromides, and tertiary alkyl bromides. Reaction conditions: under an air atmosphere, **1** (0.15 mmol), **2** (0.10 mmol), **3** (0.25 mmol), $NiBr_2$•dtbbpy (10 mol %), Zn (2.0 equiv), NaI (3.0 equiv), and THF (0.4 μL/mg) were placed in a ball-milling vessel (stainless, 1.5 mL) loaded with one grinding ball (stainless, diameter: 5.0 mm). After the vessel was closed, it was placed in the ball mill (Retsch MM500, 30 Hz, 1.0 h). Isolated yields are reported.

this protocol. When the cyano group was replaced by other functional groups, allene (**5**) could still be obtained with a moderate yield, indicating that *tert*-butyl, methoxy, trifluoromethyl, ester, formyl, acetamido, *meta*-cyano, and *para*- and *meta*-polysubstituted groups are compatible with the reaction conditions (**5a**−**5k**, with yields of 43−74%). Additionally, *tert*-alkyl bromides with both acyclic and cyclic alkyl groups were applicable to the reaction. Both acyclic and cyclic alkyl bromides provided the products in excellent yields (**6a**−**6d**, 80−94%). Unfortunately, secondary and primary alkyl bromides were unable to yield the corresponding allene products (Figure S3), and TLC indicated that most of the enyne and aryl bromides remained unreacted. This implies that it is difficult to generate the corresponding primary alkyl and secondary alkyl radicals under this ball-milling conditions.[18,20]

To further demonstrate the practicality of our method, we performed a gram-scale synthesis of compound **4a** under ambient air conditions, using bench-grade tetrahydrofuran as the LAG. This process yielded 1.65 g of compound **4a**, with a yield comparable to that of the previous small-scale synthesis (85%, Figure 3). Subsequently, a series of late-stage

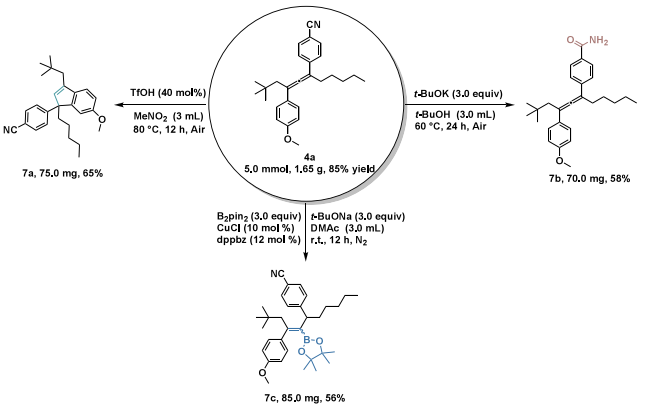


Figure 3. Scale-up reaction and synthetic applications.

functionalizations were performed on **4a** to emphasize the potential utility of the newly developed allenes. Treating allene **4a** under acidic conditions generated multisubstituted 1*H*-indene **7a** with a 65% yield. Under alkaline conditions, the hydration reaction of the cyano group proceeded smoothly, yielding allyl amide **7b** with a 58% yield. Additionally, using copper chloride and di(phenylphosphino)benzene as catalysts, allene **4a** was easily converted into vinyl-Bpin **7c**.

To gain a deeper understanding of the reductive 1,4-alkylarylation pathway, a series of mechanistic experiments were conducted (Figure 4). When 1.5 equivalents of (2,2,6,6-tetramethylpiperidin-1-yl)oxyl (TEMPO) was added (Figure 4A), the 1,4-alkylarylation of **1a** was completely inhibited, and most of the enyne and aryl bromides remained unreacted. These results unequivocally suggested the involvement of alkyl radicals in the reaction. To further elucidate the nickel catalytic cycle, the aryl-nickel (II) complex **Ni-1** was presynthesized and then reacted with **1a** and **3a** (Figure 4B). In this scenario, the desired product **4a** was obtained with an isolated yield of 17%. Furthermore, using **Ni-1** (10 mol %) as the catalyst resulted in a 71% yield of **4a**. We attribute this to the instability of aryl-nickel(II) complex **Ni-1**, which is prone to rapid decomposition during the reaction.[21] Additionally, the use of an equivalent amount of **Ni-1** was proposed to accelerate the formation of protonated products or direct coupling with alkyl radicals, which ultimately decreases the yield of the desired target compound.[22] However, when catalytic amounts of **Ni-1** are used in combination with reductants and aryl bromides, **Ni-1** can be continuously regenerated. Aryl bromides are essential for participating in the catalytic cycle, facilitating the ongoing generation of **Ni-1**.[21,22] Based on these results and previous literature,[18,23] two plausible mechanisms for this nickel-catalyzed cascade were proposed (Figure 4C). Path I: The reaction begins with the formation of a Ni^{I} species $[Ni^{I}LnX]$ (**A**). This species reacts with the tertiary alkyl bromide via a SET pathway to produce $[Ni^{II}LnX_2]$ (**B**). Concurrently, the generated tertiary alkyl radical undergoes radical addition with 1,3-enyne to form the propargyl radical **E**, which tautomerizes to yield the allenyl radical **F**. Complex **B** is reduced by zinc to generate $[Ni^{0}LnX]$ (**C**), which then undergoes oxidative addition with aryl bromides to form the Ni^{II} complex $[Ar\text{-}Ni^{II}Ln\text{-}Br]$ (**D**). The combination of **D** and **F** produces the key intermediate Ni^{III} complex **G**. Finally, the

https://doi.org/10.1021/acs.orglett.4c02729
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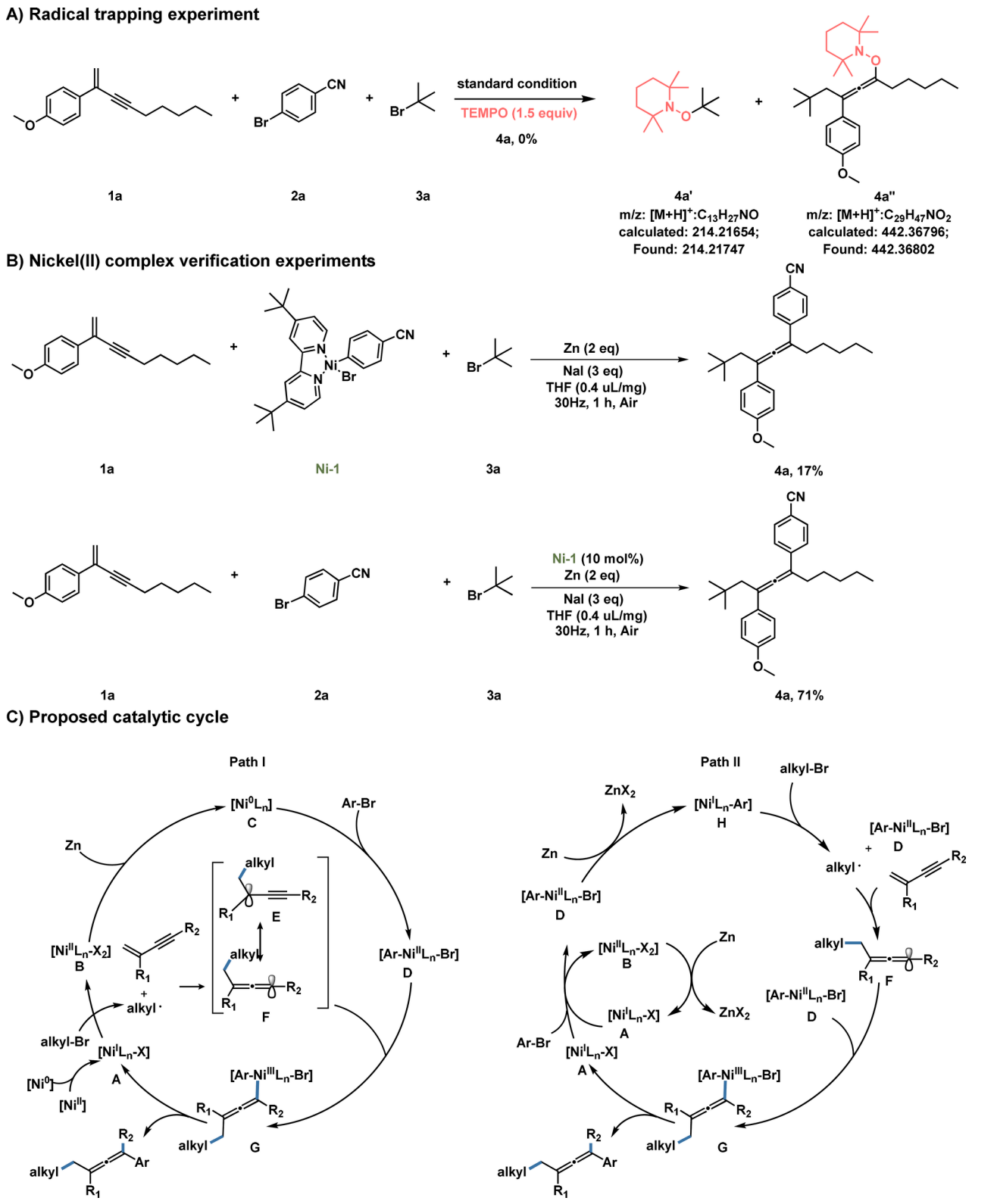


Figure 4. Mechanistic studies and proposed catalytic cycle.

reductive elimination of **G** yields the allene product and regenerates [Ni^{I}LnX] (**A**). Path II: A bimolecular oxidative addition of [Ni^{I}LnX] (**A**) with aryl bromides provides **D**, wherein complex **A** is generated by the reduction of nickel(II) complex **B** with zinc. Subsequently, reduction of **D** with zinc provides aryl-nickel(I) complex (**H**), which reacts with tertiary alkyl bromide via a SET pathway to produce **D** and the tertiary alkyl radical. Radical addition of 1,3-enyne delivers a new alkyl radical **F**. Combination of **D** with **F** provides the key intermediate **G**. Finally, reductive elimination of **G** affords the final product and regenerates **A**. These mechanistic insights highlight the role of nickel intermediates and radical species in

https://doi.org/10.1021/acs.orglett.4c02729
Org. Lett. 2024, 26, 7688−7694

the reductive 1,4-alkylarylation reaction, further validating the proposed catalytic cycle.

In summary, we have developed an efficient mechanochemical nickel-catalyzed three-component reductive 1,4-alkylarylation method for the catalytic conversion of 1,3-enynes to tetrasubstituted allenes. This method effectively addresses the challenge of selectively transforming conjugated systems, such as 1,3-enynes, which contain multiple active sites suitable for coupling reactions. Several key advantages make this method particularly attractive and practical. 1. Use of stable, readily available starting materials: The reaction employs stable, unactivated 1,3-enynes, commercially available aryl bromides, and alkyl bromides as starting materials, avoiding the need for air-sensitive or laboriously prepared reagents. This enhances both the accessibility and scalability of the reaction. 2. Simplified reaction conditions: The procedure does not require inert conditions or high temperatures, which are often essential for traditional reductive coupling reactions. This simplification reduces the overall cost and complexity of the reaction setup, making it more accessible to a wider range of chemists. 3. Rapid and efficient transformation: Under ball-milling conditions, various substrates can achieve rapid and efficient transformation within a remarkably short reaction time of just 1 h, representing a significant improvement over traditional reactions. Overall, this innovative mechanochemical approach offers a more efficient and practical method for radical reductive coupling reactions, with significant potential for diverse applications in organic synthesis.

■ ASSOCIATED CONTENT

Data Availability Statement

The data underlying this study are available in the published article and its Supporting Information.

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acs.orglett.4c02729.

Experimental details and procedures; spectra of all products (PDF)

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Author Contributions

H. L. and X. W. conceived the study; H. L., B. W., Y. Y., and S. F. designed the experiments, performed all experiments, and analyzed the data. H. L., S. W., and X. W. wrote the manuscript with input from all the other authors. X. W. is the lead contact.

Notes

The authors declare no competing financial interest.

■ ACKNOWLEDGMENTS

We thank the Natural Science Foundation of Top Talent of SZTU (grant no. GDRC202122), Featured Innovation Foundation of Ordinary Higher Education Institutions of Guangdong Province (grant no. 2021KTSCX112), High-Level Introduction Plan of Shaanxi Province (71240000000111), Natural Science Foundation of Shaanxi Province (2022JM-085, 2022JQ-121), and the start-up funds from Xi'an Jiaotong University (XJTU).

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